MIT NSE | 22.S906 Fusion Engineering
POPCON Project

Due date: TBD
Revision: 07 Oct 26

Objective

Plasma Operating CONtours (POPCONs) are utilized to identify the viable operating space for a magnetically confined plasma by including plasma performance, physics limits, and other constraints. The level of complexity and maturity can vary from very simple (simple analytical scalings, a few basic plasma limits, no administrative limits, etc.) to very advanced (computationally intense models for power density terms, numerous parameter-dependent administrative and engineering limits, etc.). The output of a POPCON, as you will hopefully see in this project, has a high degree of sensitivity on the development and inclusion of the inputs. POPCONs are typically utilized in the early scoping stage of a given tokamak design to enable rapid iterations between the physicists, engineerings, and managers responsible for converging to a point design.

The primary objectives of this project are:

1. Learn how to install, operate, and modify a professional POPCON tool
2. Understand the fundamental dependencies and sensitivities of tokamak operating spaces
3. Analyze an older tokamak design to determine if its promises hold true in a modern analysis
4. Assess the extent to which generative AI can replicate the results of a professional POPCON tool

Preparation

The assignment is composed of the follow parts:

1. Pick a tokamak to model: Identify an older, less well-analyzed tokamak from fusion's past. (SPARC and ITER have been massively analyzed with very complex tools and, so, they just aren't as interesting for us to explore). Feel free to ask for advice, but here are three suggestions:
    a. The Ignitor Tokamak
    b. The FIRE Tokamak
    c. The Vulcan Conceptual Design

2. Install a professional POPCON tool: You should identify and install an advanced, professionally maintained POPCON code on your computer. Two suggestions:
    a. OpenPOPCON: Code originating in NSE 22.63 design course that is now developed and maintained primarily by physicists at Columbia University: https://github.com/hansec/OpenPOPCON
    b. CFSPOPCON: Code tracing its origens all the way back to Bob Mumgaard's 2015 POPCON code in IDL to explore potential operating spaces for SPARC. Now maintained by CFS: https://github.com/cfs-energy/cfspopcon

Once you have installed the tool, please run some of the provided examples and ensure that (1) you can generate the correct POPCON output without errors and (2) you can modify the example source code and re-run the POPCON analysis to show changes. You should also take a quick look through the software and try to understand the basic workflow of the code.

3. Use generative AI to build your own POPCON tool: Pick your favorite AI tool, or better, as the best practices for using generative AI prescribe, pick your top two favorite AI tools so that you can compare and contrasts the results to assess the degree of trustworthiness and fidelity. Use your knowledge of magnetically confined tokamak plasmas to develop your own POPCON tool. You can decide on the level of complexity of the input, but you should ensure that you have a relatively comprehensive inclusion of the basic power terms, the plasma physics limits and even the relevant engineering and/or administrative limits based on the tokamak you selected above. Ensure that your generative AI tools can parameterize at least a few important inputs and also output a graphical POPCON that is intelligible.

Assignment

You should submit a written report that addresses the following three topics:

1. Analyze your chosen tokamak's claims: Use either the professional or your generative POPCON tool to analyze the performance claims of the tokamak that you have chosen. State clearly what the device (*e.g.* from a standard reference design paper) claims to achieve and whether or your not your POPCON analsyis supports the claim(s). Should be 2 pages or less.

2. Compare and contrast POPCON tools: For an identical set of input parameters for a single tokamak design (*i.e.* everything is identical with the exception of the machinery of the POPCON tools, assess how well your generative AI POPCON tool compared to the professional POPCON tool. Should be 2 pages or less.
    a. To what extent does your gAI POPCON tool match the professional POPCON tool?
    b. How detailed was the physics and constraints in your gAI POPCON tool?

3. Feedback on the assignment: This is a new assignment in a new course, and I am eager to incorporate feedback on how feasible this assignment was (*e.g.* Can students utilize generative AI tools to build a realistic POPCON tool? How much time and effort had to be put in to get meaningful results?) and how insightful this was as a learning exercise to reinforce what we did in class. Should be 1 page or less.

NOTE: You may not use generative AI for any part of the writing assignment, as ideating, drafting, and polishing technical writing is an essential human development skill that is inappropriate to cognitively offload to AI in an educational setting. Using AI tools will result in 50% reduction of credit.